

ISIT312/ISIT912 Big Data Management

MapReduce Framework

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MapReduce Framework

Outline

MapReduce

Real world scenario: log data analysis

MapReduce implementation in Hadoop

MapReduce

MapReduce is the most important processing framework in **Hadoop**

Many high-level data processing languages are abstractions of MapReduce, for example **Pig**, **Hive** and **Spark** are heavily influenced by **MapReduce** concepts

Historically, **Hadoop** version 1 supported **MapReduce** only

MapReduce is also a platform and **language-independent programming model** at the heart of most big data and NoSQL platforms

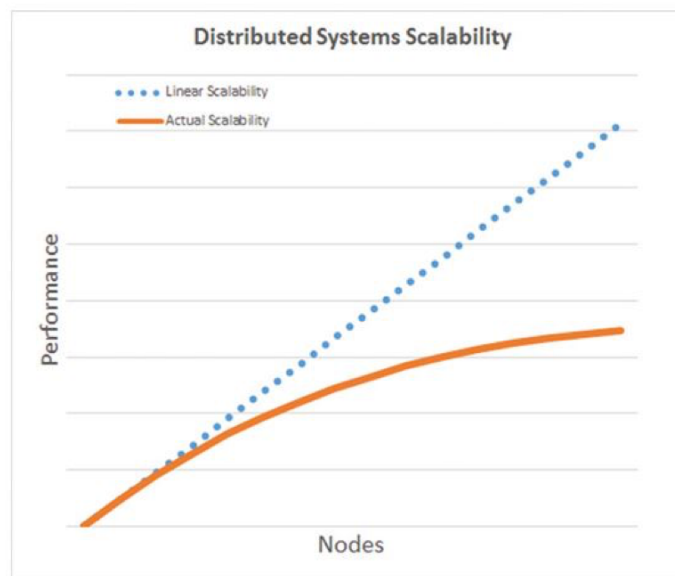
A **programming model** is a pattern/format in accordance to which we write our programs

The logic of a **MapReduce** application consists of a **Map** phase and a **Reduce** phase

MapReduce

Limitations of early distributed computing and grid computing frameworks:

- Complexity in parallel programming
- Hardware failures
- Bottlenecks in data exchange
- Scalability problem



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MapReduce

The 2004 Google MapReduce white papers determined the following design goals of MapReduce

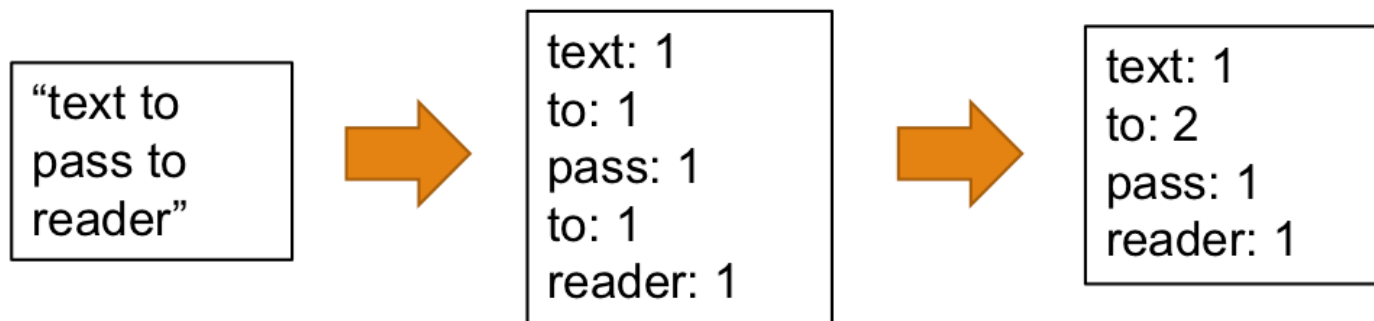
- Automatic parallelization and distribution
- Fault tolerance
- Input/output (I/O) scheduling
- Status and monitoring

MapReduce

MapReduce model uses **key-value** pairs for processing data

Key	Value
City	Sydney
Date	[02-03-2017, 02-04-2016]

WordCount: MapReduce Hello World example



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A Real-World scenario: log data analysis

In online purchasing, users sometimes abandon their shopping carts before completing the purchase

In order to improve their business, companies are usually interested to find out more about the nature of these abandoned purchases

A **MapReduce** job for this analysis

1. The final pages visited by the users
2. The contents of the abandoned shopping carts
3. The user session's transaction state

Map phase

1. aggregated data for the total number of abandoned carts
2. the most common final page visited by the users when they ended their website visit, abandoning their shopping carts.

Reduce phase

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MapReduce implementation in Hadoop

Hadoop MapReduce frees the users from the low-level communication and coordination of nodes and processes

Let programmers focus on the **MapReduce** implementation and a few configuration parameters

As the data file is usually too large to be stored in a single persistent storage device of the commodity hardware, **Hadoop** handles the shipment of code to data fragments (**data locality**)

This can dramatically reduce the overhead of network transmits

MapReduce implementation in Hadoop

Why **Hadoop** is useful to Big Data ?

- Cost-effective fault-tolerant storage (HDFS)
- Scalability
- Data that is ingested may be interpreted at runtime
- Low cost in storing unstructured and semi-structured data
- Fast transfer of data into storage
- Separation of programming logic and scheduling/management
- Multiple levels of distributed system abstractions: **Hive**, **Pig**, **Spark**
- Multi-language tooling: Java, Python: **MapReduce**; SQL: **Hive**; data-flow: **Pig**; Scala, Python: **Spark**;

References

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[Apache Hadoop](#)

[Apache Hadoop, MapReduce Tutorial](#)